

PyTodo CLI

The ultimate, high-urgency, retro-terminal to-do system designed to vaporize procrastination through daily sprint mechanics, loss aversion, and cross-device telepathy.

```
[16:45:00] [TODAY REMAINING: 07h 14m 59s] [TASKS: 3/5 DONE] [● SYNCED]
pytodo> add Launch epic product by 6pm for 45m
✓ [CREATED] #1 "Launch epic product" [DUE: 18:00 (1h 15m left)] [EST: 45m]
♪ 🎵 Upbeat 8-bit victory arcade chime echoes in your headphones ...
```

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Let's be completely honest: traditional to-do apps are **digital graveyards for forgotten ambitions**. You create a task like *"Learn Japanese"* or *"Organize garage"*, it sits there for 9 months, rolls over automatically every morning, and teaches your brain that deadlines are imaginary suggestions. Zero consequence. Zero urgency. Total guilt.

The Stale Graveyard (Other Apps)

Tasks roll over infinitely into tomorrow. You wake up to 87 red overdue badges. You feel overwhelmed, close the app in disgust, and watch YouTube cat videos instead.

The High-Stakes Daily Sprint (PyTodo)

Every day is a closed, high-intensity tournament. At midnight (00:00), unfinished tasks are wiped from your radar into the Graveyard. You start every morning with a fresh, clean slate!

02 The 60-Second Quickstart: Level 1 Novice

HANDS-ON TUTORIAL

You can run PyTodo in two awesome ways: **1) Pure Terminal** locally via Python, or **2) Instant Browser PWA** (powered by WebAssembly Pyodide + xterm.js) with zero installations required!

```
bash / terminal - 60-Second First Run

# 1. Fire up the CLI (or launch browser at localhost:8000)
~/project $ python main.py ls
NO TASKS FOUND FOR TODAY (2026-09-23)
Your slate is pristine! Add a mission with: add <title> |

# 2. Add your very first quest (Natural language auto-detects deadline & duration!)
~/project $ python main.py add "Defeat procrastination dragon" by 5pm for 30m
✓ [CREATED] #1 Defeat procrastination dragon [Due 17:00] [30m]

# 3. Check your active radar
~/project $ python main.py ls
ID STATUS SYNC DEADLINE REMAINING TASK
-----
1 [TODO] ✓ 17:00 01h 15m Defeat procrastination dragon (30m)

# 4. Slay the quest and claim victory!
~/project $ python main.py done 1
★ [VICTORY] Task #1 completed! [1/1 done - 100% Daily Perfection]
```



Pro-Tip for Terminal Speedrunners

Inside the interactive web terminal, you don't even need to type `python main.py`. Just type commands directly: `add ...`, `ls`, `done 1`. Or use single-letter aliases: `a`, `l`, `d`!

Forget memorizing complex POSIX flag gymnastics. PyTodo includes a built-in **Natural Language Parser** that understands how humans actually speak and think about time. Just speak your mind, and PyTodo automatically extracts the deadline and duration!

WHAT YOU TYPE	AUTO-EXTRACTED PROPERTIES	WHY IT'S AWESOME
<code>add Finish report by 5pm for 45m</code>	<code>DUE: 17:00</code> <code>DURATION: 45M</code>	Sets both deadline and target work sprint effortlessly.
<code>add Team standup at 2:30pm</code>	<code>DUE: 14:30</code>	Recognizes 12-hour AM/PM formats automatically.
<code>add Fix production bug in 45m</code>	<code>RELATIVE DUE: NOW + 45M</code>	Calculates exact target wall-clock time on the fly.
<code>add Deep work session for 2h</code>	<code>DURATION: 120M</code>	Understands hours (<code>h</code>) and minutes (<code>m</code>).
<code>add "Audit code" --due 18:00 --duration 1h</code>	<code>DUE: 18:00</code> <code>DURATION: 60M</code>	Explicit POSIX flags are still 100% supported!

Token Isolation Technology

PyTodo safely extracts time keywords from the end of the sentence without mangling your title. So *"Meeting at 2pm"* cleanly becomes title *"Meeting"* with deadline `14:00` !

Precedence Hierarchy

If you pass both natural language and an explicit flag (e.g. `add "Sync at 2pm" --due 15:00`), explicit POSIX flags always take precedence. You retain 100% surgical control.

Natural Language in Action

```
pytodo> a Ship v2 release by 4pm for 1h
✓ [CREATED] #2 Ship v2 release [Due: 16:00] [Duration: 60m]
Notice how alias 'a' saved you typing 2 characters. Speedrunners rejoice!
```



The 3-Second Add Challenge

Try adding 3 missions right now without touching your mouse. Type `a Gym in 1h`, then `a Lunch at 1pm`, then `a Read for 30m`. In under 9 seconds, your whole day's battle plan is set!

Running `ls` isn't just dumping a list—it's reading your tactical mission radar. PyTodo dynamically computes the remaining time for every task and grades them with an **adaptive color urgency heatmap**:

CALM GREEN > 4h Left

Relaxed zone. You have plenty of runway. Ideal for thoughtful design, deep focus, and strategic planning.

WARNING AMBER < 2h Left

The clock is ticking! Wrap up conversations, close social media tabs, and enter high-gear execution.

CRITICAL RED < 45m Left

Defcon 1! Blinks bold in your terminal! Drop everything else and push this task over the finish line.

```
●●● The Live Ticking Header & Visual Radar: ls

[TODAY REMAINING: 04h 12m 30s] [TASKS: 2/4 DONE] [● SYNCED] [STREAK: 5 🔥]

ID STATUS SYNC DEADLINE REMAINING TASK
1 [DONE] ✓ 11:00 --- Draft architecture spec
2 [DONE] ✓ 13:30 --- Deploy microservice to staging
3 [TODO] ✓ 18:00 01h 45m Review pull requests for release (45m)
4 [TODO] ✓ 16:45 00h 30m Submit quarterly tax paperwork (Due Soon!)

💡 Tip: Type 'top' to open the live auto-refreshing full-screen cockpit!
```

 Intelligent Screen-Width Adaptation

PyTodo measures your terminal column width on every render. If you resize your browser window or view on a mobile phone (less than 65 columns), PyTodo automatically strips decorative padding, wraps titles safely, and renders a sleek, ultra-compact view that never breaks line geometry!

Life is fluid. Priorities shift. PyTodo equips you with surgical in-place editing tools so you can update, re-time, or discard missions in milliseconds without losing your momentum.

✓ Marking Complete (`done` / `d`)

Crush a task with `done 1` (or `d 1`). PyTodo plays an upbeat 8-bit retro fanfare and tallies your daily completion rate.

Works on subtasks too: `done 1.2` marks subtask 2 of task 1 complete!

↺ Reverting Mistakes (`undone` / `u`)

Accidentally hit complete on something you didn't actually finish? Zero sweat.

Type `undone 1` (or `u 1`) to immediately restore the mission back to active status.

✏ In-Place Surgical Editing (`edit` / `e`)

• Rename: `edit 1 --title "Better Mission Name"`

• Push Deadline: `edit 1 --due 19:30`

• Clear Deadline: `edit 1 --clear-due`

• Change Duration: `edit 1 --duration 45m`

🗑 Vaporizing Clutter (`rm` / `del`)

If a task becomes totally obsolete, delete it cleanly with `rm 1` or `del 1`.

Want to remove just a subtask? Run `rm 1.3` to cleanly excise only that checkpoint.

Editing & Completion Example

```
pytodo> edit 3 --due 20:00 --duration 90m
```

```
✓ [UPDATED] Task #3 updated: Due 20:00, Est. 90m.
```

```
pytodo> done 3
```

```
★ [VICTORY] Task #3 marked complete! [3/4 done today - 75%]
```

Big goals trigger procrastination when lump-summed into an intimidating block like *"Build entire backend"*. PyTodo lets you break any mission down into **checkpoint subtasks** with their own numbering, deadlines, and checkboxes!

```
● ● ● Subtask Tree Hierarchy Demonstration

# Add subtasks using the parent task ID (Natural language works here too!)
pytodo> subtask 1 Gather user analytics by 3pm for 20m
pytodo> subtask 1 Draft presentation slides by 4:30pm for 45m
pytodo> subtask 1 Practice dry run delivery for 15m

# View the nested tree structure in 'ls'
pytodo> ls
ID STATUS SYNC DEADLINE TASK
1 [TODO] ✓ 17:00 Prepare Q3 Product Launch [1/3 done]
├── 1.1 [DONE] ✓ 15:00 Gather user analytics (20m)
├── 1.2 [TODO] ✓ 16:30 Draft presentation slides (45m)
└── 1.3 [TODO] ✓ --- Practice dry run delivery (15m)

# Knock out subtasks individually!
pytodo> done 1.2
✓ Subtask 1.2 completed! Parent progress updated to [2/3 done].
```

👑 The Auto-Cascade Rule

When you complete all subtasks and run `done 1`, PyTodo automatically marks any lingering subtasks complete and announces the entire parent quest conquered!

⚡ Speedrunner Shortcut

You can add subtasks simultaneously when creating a task: `add "Quarterly Plan" -s "Financials" -s "Hiring goals"`. Instantly creates parent and children in one stroke!



Rule of Micro-Steps

Never leave a task in `[TODO]` status if you feel resistance. Attach a 5-minute subtask like `subtask 1 Open blank document`. Once that tiny first checkpoint is checked off, momentum takes over!

Hyperfocus Pomodoro Mode: Your Personal Boss Battle

COMMAND: FOCUS (ALIAS: F)

Distractions kill productivity. PyTodo's **Hyperfocus Mode** locks your terminal onto a single mission. It dims peripheral noise, displays a ticking ASCII progress bar, and keeps you immersed until the timer completes.

Space Pause

Need coffee or answering a door? Tap space to pause timer instantly.

+ / - Time

Add or remove 5 minutes on the fly if your sprint needs extra runway.

d Victory

Finished early? Press **d** to complete the task with fanfare and exit.

q Quit

Safely exits focus mode
back to the normal
prompt anytime.



Synthesized Victory Alarm

When the countdown reaches **00:00**, PyTodo rings an upbeat retro 8-bit audio fanfare to celebrate your focused sprint. Stretch your shoulders, grab a glass of water, and conquer the next quest!

Inspired by Linux's legendary `htop` utility, running `top` (or `watch`) transforms your terminal into a live, self-updating mission control center. Keep it open on a vertical or secondary monitor to stay continuously anchored!

```
● ● ● Live Mission Control Dashboard (top / watch)
```

```
PYTODO MISSION CONTROL v2.4 | LOCAL TIME: 16:45:12 | SYNC: ONLINE [●]
```

```
DAYLIGHT PROGRESS : [██████████] 69.8% (07h 14m left until wipe)
```

```
TASK COMPLETION   : [██████████] 75.0% (3/4 missions completed)
```

```
ACTIVE COUNTDOWNS:
```

```
#4 Submit tax paperwork 🕒 00h 29m 48s [CRITICAL PRIORITY]
```

```
(Press 'q' or Ctrl+C to return to standard command prompt)
```

Dual Progress Bar Telemetry

The top bar tracks how much of the daylight hours have elapsed. The bottom bar tracks your task completion percentage. When the green completion bar outpaces the daylight bar, you are winning the day!

Full-Screen Cinema Mode

Want an even cleaner setup? Run `top -full` (or `t -full`). PyTodo expands into a borderless, focused monitoring kiosk that looks like something out of NASA headquarters.



Pro Tip: The Ambient Motivation Display

Keep **top** running on an old iPad or tablet on your desk. Watching the real-time clock tick down creates healthy, gentle accountability that keeps you completely on track without intrusive notifications.

What happens when the clock strikes midnight? **The Midnight Wipe activates**. Uncompleted tasks vanish from your active board so you are never paralyzed by accumulated debt. But they aren't deleted into oblivion—they are safely archived in the **Graveyard**!

The Graveyard Ledger (`history`)

Type `history` (or `overdue` / `graveyard`) to inspect uncompleted missions from the past 1 to 7 days.

Each task receives a unique historical index (e.g. `H1`, `H2`), showing its expiration timestamp.

The Lazarus Pit: Resurrection (`revive`)

Life happens. If an expired task is still important today, give it a second chance!

Type `revive H1` (or `rv H1`). PyTodo immediately restores the task into today's active sprint!

Inspecting & Resurrecting Overdue Tasks

```
pytodo> history
```

```
OVERDUE GRAVEYARD (PAST 7 DAYS RETENTION)
```

```
TAG DATE EXPIRED TASK TITLE SUBTASKS |
```

```
H1 2026-09-22 Finalize contract legal clauses [1/2 done] |
```

```
H2 2026-09-21 Clean up old Docker images None |
```

Resurrect any task into today's sprint using: `revive <tag>` (e.g. `revive H1`)

```
pytodo> revive H1
```

```
🌟 [RESURRECTED] "Finalize contract legal clauses" has been restored as Task #5 for today!
```

Customizing History Retention

By default, PyTodo preserves overdue tasks for 7 days. Prefer ultra-hardcore minimalism? Run `config retention 1` to retain only yesterday's tasks, or `config retention 3` for a 3-day buffer.

Consistency builds empires. PyTodo tracks your **Discipline Streak** based on consecutive days where 100% of your daily commitments were conquered before the midnight bell. No half-measures!

```
Discipline Streak & 7-Day Completion Ledger

pytodo> streak
🔥 CURRENT DISCIPLINE STREAK: 6 CONSECUTIVE DAYS OF 100% COMPLETION!
🏆 LONGEST STREAK RECORD : 14 DAYS

RECENT 7-DAY BATTLE LEDGER:
2026-09-17 (Thu): [●●●●●] 5/5 Done (100%) - VICTORY
2026-09-18 (Fri): [●●●●] 4/4 Done (100%) - VICTORY
2026-09-19 (Sat): [●●●] 3/3 Done (100%) - VICTORY
2026-09-20 (Sun): [●●] 2/2 Done (100%) - VICTORY
2026-09-21 (Mon): [●●●●●●] 6/6 Done (100%) - VICTORY
2026-09-22 (Tue): [●●●●] 4/4 Done (100%) - VICTORY
2026-09-23 (TODAY): [●●●📅] 3/4 Done (In Progress – Complete by midnight to reach 7! 🔥)
```

🧠 The Psychology of Loss Aversion

Behavioral science shows we hate losing something twice as much as we enjoy gaining it. Once you reach a 7-day or 14-day streak, you will find unexpected energy at 11:30 PM to finish your last task rather than see your streak reset to 0!

🛡️ Smart Rest Days

Taking Sunday off? If you have 0 tasks on a given calendar date, PyTodo preserves your active streak rather than penalizing you. Only days with active tasks are evaluated for 100% completion!

🔥 The 7-Day Challenge

Can you maintain a 100% streak for 7 consecutive days? Commit to just 3 to 5 realistic tasks per day, conquer them before midnight, and watch your daily productivity skyrocket!

Aesthetics inspire focus. PyTodo includes **7 handcrafted retro color schemes**. Switch palettes instantly with `theme <name>` without restarting the app:

THEME NAME	VISUAL PROFILE & AESTHETICS	ACTIVATION COMMAND
Classic	Authentic vintage phosphor green and amber CRT vibes.	<code>theme classic</code>
Matrix	Deep obsidian background with radiant neon green digital rain.	<code>theme matrix</code>
Cyberpunk	High-voltage neon magenta, electric cyan, and deep midnight navy.	<code>theme cyberpunk</code>
Dracula	Gothic vampire purple, soft pastel pink, and rich charcoal gray.	<code>theme dracula</code>
Nord	Calm, frosty Arctic blues, cool slate grays, and icy whites.	<code>theme nord</code>
Monokai	Sublime Text classic warm yellows, vibrant magentas, and bright greens.	<code>theme monokai</code>
Solarized	Low-glare warm designer earth tones engineered for night reading.	<code>theme solarized</code>

Switching Themes in Real Time

```
pytodo> theme cyberpunk
🎨 [THEME UPDATED] Activated theme: 'cyberpunk' (Neon Magenta & Electric Cyan)
Changes are saved instantly to local storage and sync across your linked devices!
```



Mood-Based Theme Strategy

Try using **Nord** or **Solarized** during daytime work sessions for crystal-clear readability, and switch to **Cyberpunk** or **Matrix** for late-night hacking sprints!

Nobody wants to remember yet another email and 16-character password just to check off daily tasks. PyTodo eliminates account fatigue with **frictionless 6-digit pairing keys**!

Device 1: Reveal Your Key

Run `code` or `pair new`. PyTodo gives you a unique 6-digit pairing code (e.g. `849201`).

This key represents your private encrypted channel on Supabase cloud storage.

Device 2: Connect Instantly

Open PyTodo on your phone or laptop browser and run: `link 849201`.

Boom! Your tasks, subtasks, themes, and streaks sync across devices in under 200 milliseconds!

Pairing & Syncing Commands

```
pytodo> code
```

```
🔑 [PAIRING KEY]: 849201
```

```
Use 'link 849201' on your other device to connect instantly without passwords.
```

```
pytodo> sync
```

```
✓ [CLOUD SYNC COMPLETE] Local database is 100% in sync with cloud (0 pending mutations).
```

Offline-First Optimistic Speed

PyTodo writes every command locally in **0ms** to browser storage before making any network request. You can use it on a plane or in an underground subway. The moment you reconnect to Wi-Fi, pending changes automatically flush to the cloud in the background!

Productivity should feel rewarding! PyTodo includes a zero-dependency **synthesized Web Audio engine** that produces nostalgic 8-bit arcade sounds using pure oscillator waveforms right in your browser:

Victory Chime

An upbeat, ascending retro arpeggio played whenever you complete a task or subtask.

Focus Alarm

A triumphant fanfare signaling the successful conclusion of a Pomodoro sprint.

Audio Controls

Toggle sound anytime: `sound on`, `sound off`, or test audio with `sound test`.

14 Total Data Freedom: Backup & Portability

COMMANDS: EXPORT, IMPORT, CONFIG

Your data belongs to you. No proprietary lock-in. Export clean JSON snapshots or restore previous backups with a single command:

Instant JSON Export (``export``)

Type `export pretty` to view or download a human-readable JSON snapshot of your entire database.

Includes tasks, subtask hierarchies, streak records, and custom settings.

Flexible JSON Import (``import``)

- `import merge <json>`: Merges backup tasks into today's board without overwriting existing tasks.
- `import replace <json>`: Performs a pristine, fresh restoration from your snapshot.



The Ultimate Anti-Lock-In Guarantee

Because PyTodo stores standard JSON, you can pipe your export into any external tool, script, or SQLite database anytime you want. Zero lock-in. Ever.

Keep this page pinned to your monitor or desk! Here is the complete cross-reference matrix of every command, single-letter alias, and argument in PyTodo CLI:

COMMAND	ALIAS	SYNTAX & EXAMPLE	ACTION PERFORMED
<code>add</code>	<code>a</code>	<code>add "Ship feature" by 5pm for 45m</code>	Creates task with natural language parsing.
<code>ls</code>	<code>l</code>	<code>ls</code> or <code>today</code>	Shows today's active tasks with urgency heatmap.
<code>done</code>	<code>d</code>	<code>done 1</code> or <code>done 1.2</code>	Marks task or subtask complete (+ 8-bit audio).
<code>undone</code>	<code>u</code>	<code>undone 1</code>	Reverts task back to incomplete state.
<code>subtask</code>	<code>s</code>	<code>subtask 1 Step 1 by 3pm</code>	Attaches a checkpoint subtask to parent.
<code>focus</code>	<code>f</code>	<code>focus 1 -time 25m</code>	Launches Pomodoro focus screen (Space+/d/q).
<code>top</code>	<code>t</code>	<code>top</code> or <code>watch</code>	Opens live auto-refreshing monitor cockpit.
<code>edit</code>	<code>e</code>	<code>edit 1 --due 18:00 --title "... "</code>	Modifies task, subtasks, or deadlines in-place.
<code>rm</code>	---	<code>rm 1</code> or <code>del 1</code>	Deletes task or subtask cleanly.
<code>history</code>	---	<code>history</code> or <code>graveyard</code>	Inspects past uncompleted tasks (1–7 days).
<code>revive</code>	<code>rv</code>	<code>revive H1</code>	Resurrects overdue task into today's sprint.
<code>streak</code>	---	<code>streak</code>	Displays consecutive 100% completion ledger.
<code>theme</code>	---	<code>theme cyberpunk</code>	Changes color palette (7 retro themes).
<code>pair</code>	<code>p</code>	<code>pair link 849201</code> or <code>code</code>	Pairs and syncs devices via 6-digit code.
<code>sound</code>	---	<code>sound on off test</code>	Controls 8-bit synthesized arcade sound FX.
<code>export</code>	---	<code>export pretty</code>	Exports JSON backup snapshot.
<code>help</code>	<code>h</code>	<code>help</code>	Displays built-in quick command help guide.



Go Forth and Conquer!

Remember: Every morning brings a clean slate. Set 3 to 5 high-impact quests, break them down with subtasks, enter hyperfocus, and conquer your day before the midnight bell tolls. Happy executing!